

Supplementary Material

Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis

This supplement lists the parameters of the synthetic catalogues used in the worked example of the main paper (Table S1) and collects additional screen captures of the *Catalogue of Catalogues* (CofC) platform that support, but are not essential to, the argument of the main paper. All captures are from the release described in the main paper, running in a browser against a local instance.

Data. Every capture uses a reduced-scale version of the worked example’s synthetic pair, built by the same seeded generator (`paper/figures/synthetic_catalogues.py`, seed 42) with each true earthquake kept with probability 1/23: a 4,922-event “GeoNet-like” catalogue and a 4,411-event “Agency B” catalogue, each uploaded as a QuakeML 1.2 file written by `python paper/figures/generate_figures.py --screenshot-catalogues`. The values quoted in the captions are those of this reduced pair, computed by the platform’s own code (`paper/figures/screenshot_expectations.ts`, which parses the files with the upload parser and drives the Analysis page’s analysis worker); the worked example of the main paper uses the full-scale pair. No real event data are shown anywhere in this supplement.

Synthetic catalogue construction

Table S1: Inputs of the synthetic worked example. Everything else the main paper reports about it (catalogue sizes, overlap, quality scores, duplicates, M_c , b -values and declustering counts) is computed from the resulting catalogues by the platform.

Component	Specification
Period and domain	2020-01-01 to 2024-12-31 (1827 days); true epicentres in 166°–179°E, 47.5°–34.5°S.
Background	10^6 independent events, uniform in time, magnitudes from a Gutenberg-Richter law with $b = 1.00$ (continuous, $M0.5$ –7.5), about 3200 events of $M \geq 2.3$ per year.
Source zones	Share of the background, centre, and Gaussian spread along / across strike (strike azimuth): Hikurangi margin 19%, (177.3°E, 40.0°S), 170/45 km (35°), depth 8–30 km; central North Island crust 22%, (176.0°E, 39.0°S), 150/60 km (40°), 2–15 km; North Island slab 16%, (175.8°E, 39.6°S), 190/60 km (40°), 40 km plus an exponential tail of scale 30 km (at most 250 km); Marlborough–Canterbury 16%, (173.2°E, 42.4°S), 130/60 km (50°), 3–20 km; Alpine Fault 14%, (170.5°E, 43.6°S), 180/50 km (55°), 2–15 km; Fiordland–Puysegur 8%, (166.9°E, 45.6°S), 110/45 km (25°), 5 km plus an exponential tail of scale 25 km (at most 150 km); distributed offshore 5%, uniform over the domain, 5–30 km.
Aftershocks	Sequences on every background event of $M \geq 4.5$, sized so that 23% of the true events of $M \geq 2.0$ are aftershocks and shared between sequences in proportion to $10^{0.8(M-4.5)}$ (Utsu, 1970); times from the modified Omori law ($p = 1.1$, $c = 0.05$ d, up to 365 d; Omori, 1894; Utsu, 1961); epicentres uniform in a disc of radius twice the rupture length $\log_{10} L = 0.59M - 2.44$ (Wells and Coppersmith, 1994); depth of the mainshock ± 3 km; magnitudes from a law with $b = 1.30$ truncated 1.2 units below the mainshock (Båth, 1965).
Station networks	Jittered grids on New Zealand land (Natural Earth 1:10m): GeoNet-like, $0.36^\circ \times 0.47^\circ$ (~ 40 km), 166 stations over the whole country; Agency B, $0.27^\circ \times 0.35^\circ$ (~ 30 km), 67 stations in 175.5°–178.6°E, 41.4°–37.6°S.
Recording	A station records an earthquake if its hypocentral distance is below $R(M) = 10^{1.11+0.38M}$ km times a log-normal station factor ($\sigma_{\ln} = 0.25$); an agency catalogues every earthquake that at least four of its stations record. Each recording station adds a P phase, and an S phase with probability 0.6.

Component	Specification
Reported metadata	From the recording stations: used-station and phase counts, azimuthal gap (largest gap between their azimuths), nearest-station distance $\Delta_{\min}$. Horizontal uncertainty $\sigma_h = 0.9\sqrt{6/n}e^{\max(0, \text{gap}-150^\circ)/70^\circ}(1 + \Delta_{\min}/80 \text{ km})\epsilon$ km; depth uncertainty $\sigma_z = 1.6\sigma_h(1 + \Delta_{\min}/(z + 5 \text{ km}))\epsilon$, with the depth fixed (operator assigned: 12 km GeoNet-like, 10 km Agency B) and no depth uncertainty reported where $\sigma_z > 20$ km; origin-time uncertainty $0.05 + \epsilon\sigma_h/7$ s; RMS residual with median 0.20 s (GeoNet-like) or 0.14 s (Agency B), times $(1 + \Delta_{\min}/200 \text{ km})$; magnitude stations $n_{\text{amp}} \approx 0.6\text{--}0.95n$ and magnitude uncertainty $0.05 + 0.30/\sqrt{n_{\text{amp}}}$. Each ϵ is an independent log-normal factor ($\sigma_{\ln} = 0.25$; 0.2 for the time uncertainty), and the RMS residual varies log-normally about its median with $\sigma_{\ln} = 0.3$.
Reported values	True epicentre, depth, origin time and magnitude plus errors of exactly the reported size; M_L , evaluation mode manual and status reviewed for both agencies; magnitudes to full precision (GeoNet-like) or to 0.1 (Agency B).

Statistical analysis in the browser

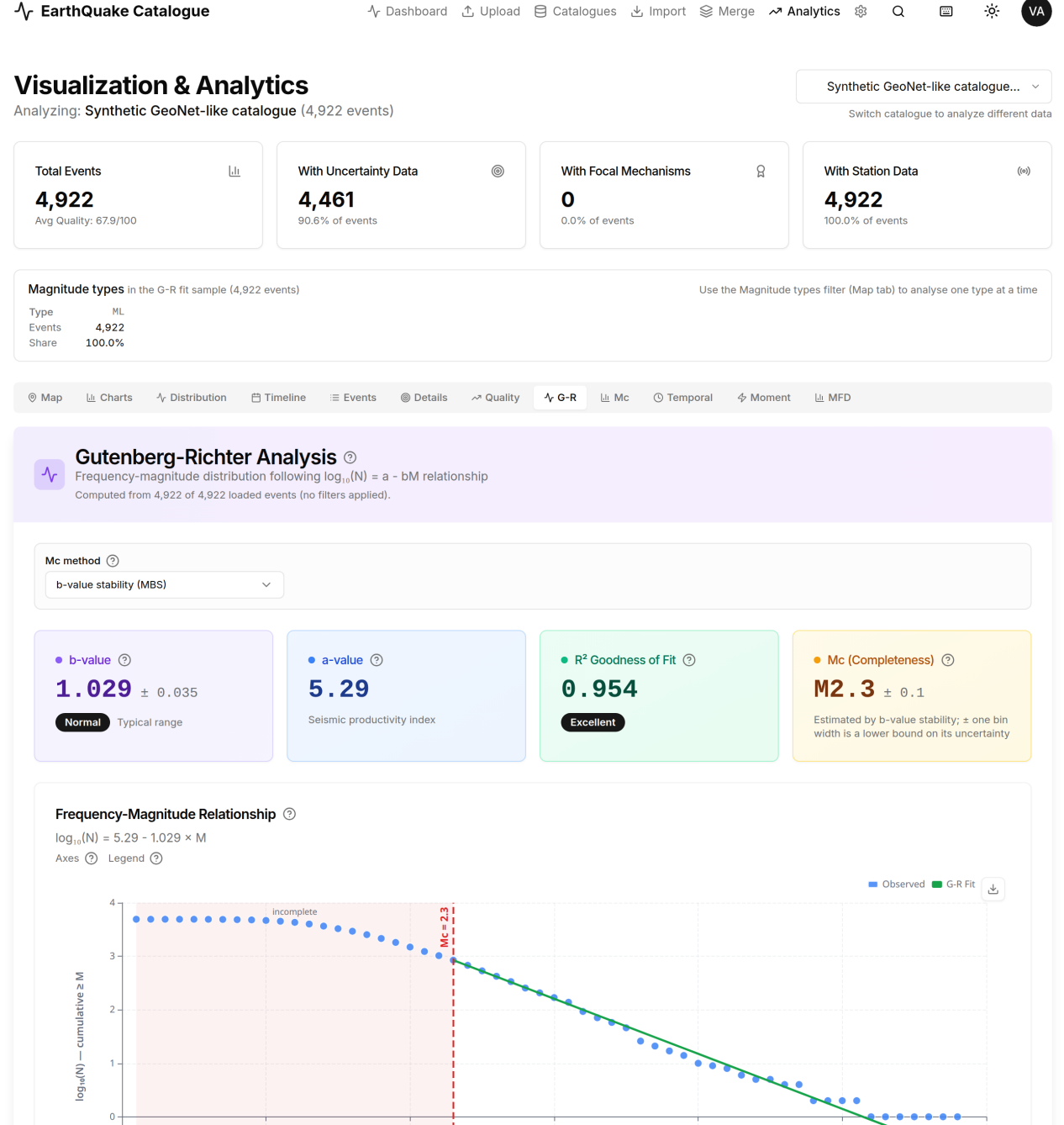


Figure S1: Gutenberg-Richter analysis of the reduced GeoNet-like catalogue, computed client-side with the default M_c settings (b -value stability, which falls back to the goodness-of-fit test and then to maximum curvature plus 0.2). The platform reports $\hat{b}$ with its counting-statistics uncertainty, the a -value, the goodness of fit and M_c with the method that produced it; the frequency-magnitude plot shades the range below M_c and overlays the fitted relation. Above $M_c = M2.3 \pm 0.1$ the fit gives $\hat{b} = 1.029 \pm 0.035$ ($N = 845$; $a = 5.29$, $R^2 = 0.954$), within its uncertainty of the $b = 1.00$ planted in the synthetic background (the sample also holds aftershocks planted with $b = 1.30$). With 845 events that uncertainty is wide, and on the full-scale catalogue of the main paper the same method places M_c higher (Figure S2). The magnitudes are reported to full precision, so no binning correction is applied. The capture is a check of the platform's behaviour on a controlled input, not a measurement of New Zealand seismicity.

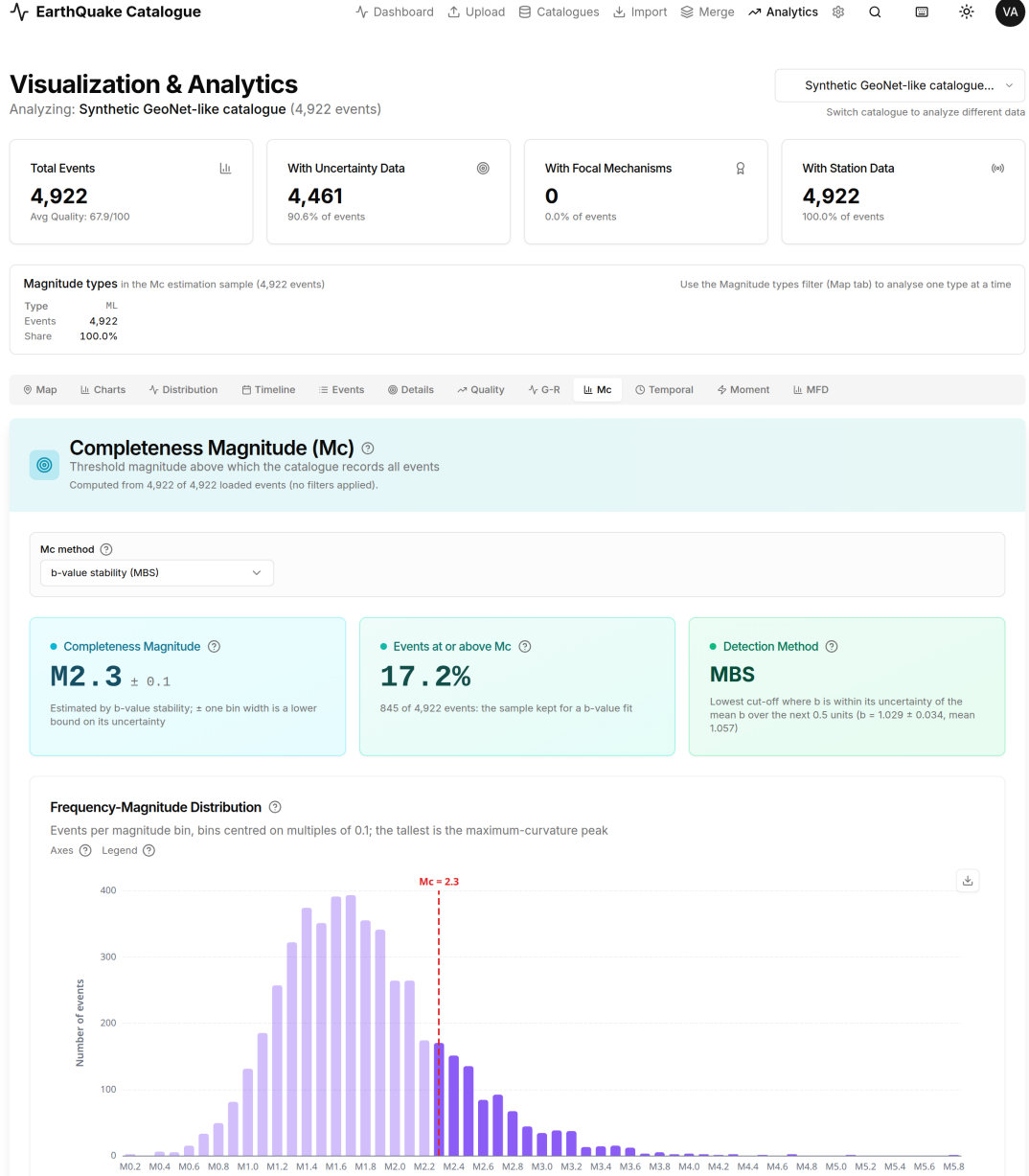


Figure S2: Magnitude-of-completeness panel for the same catalogue, with the default method, b -value stability (the main paper’s M_c section; Woessner and Wiemer, 2005); the goodness-of-fit test and maximum curvature, with a correction from 0 to 0.5, are the alternatives. $M_c = M2.3 \pm 0.1$ (one bin width, a lower bound on its uncertainty) is the lowest cut-off at which $\hat{b}$ lies within its Shi and Bolt (1982) uncertainty of its mean over the cut-offs $M2.3$ – $M2.8$: the method card quotes $\hat{b} = 1.029$, $\delta b = 0.034$ and a mean of 1.057. At or above M_c lie 17.2% of the events (845 of 4,922), the sample a b -value fit keeps; the share is not a completeness score. The distribution is drawn in bins centred on multiples of 0.1, and the fullest bin of the non-cumulative distribution is $M1.7$, so maximum curvature with its default $+0.2$ correction would give $M1.9$, and the goodness-of-fit test gives $M2.2$ at the 95% level. Below the distribution the tab plots the stability test itself, $\hat{b} \pm \delta b$ and the mean over the next 0.5 units at every cut-off with at least 50 events above it, whichever method set M_c . b -value stability is conservative on large samples, because δb falls as $1/\sqrt{N}$: on the full-scale GeoNet-like catalogue of the main paper, 23 times larger, it gives $M2.7$. The estimate is shown with its method and the values behind it, so that the completeness assumption behind any subsequent b -value is visible at the point of use.

Event-level quality index

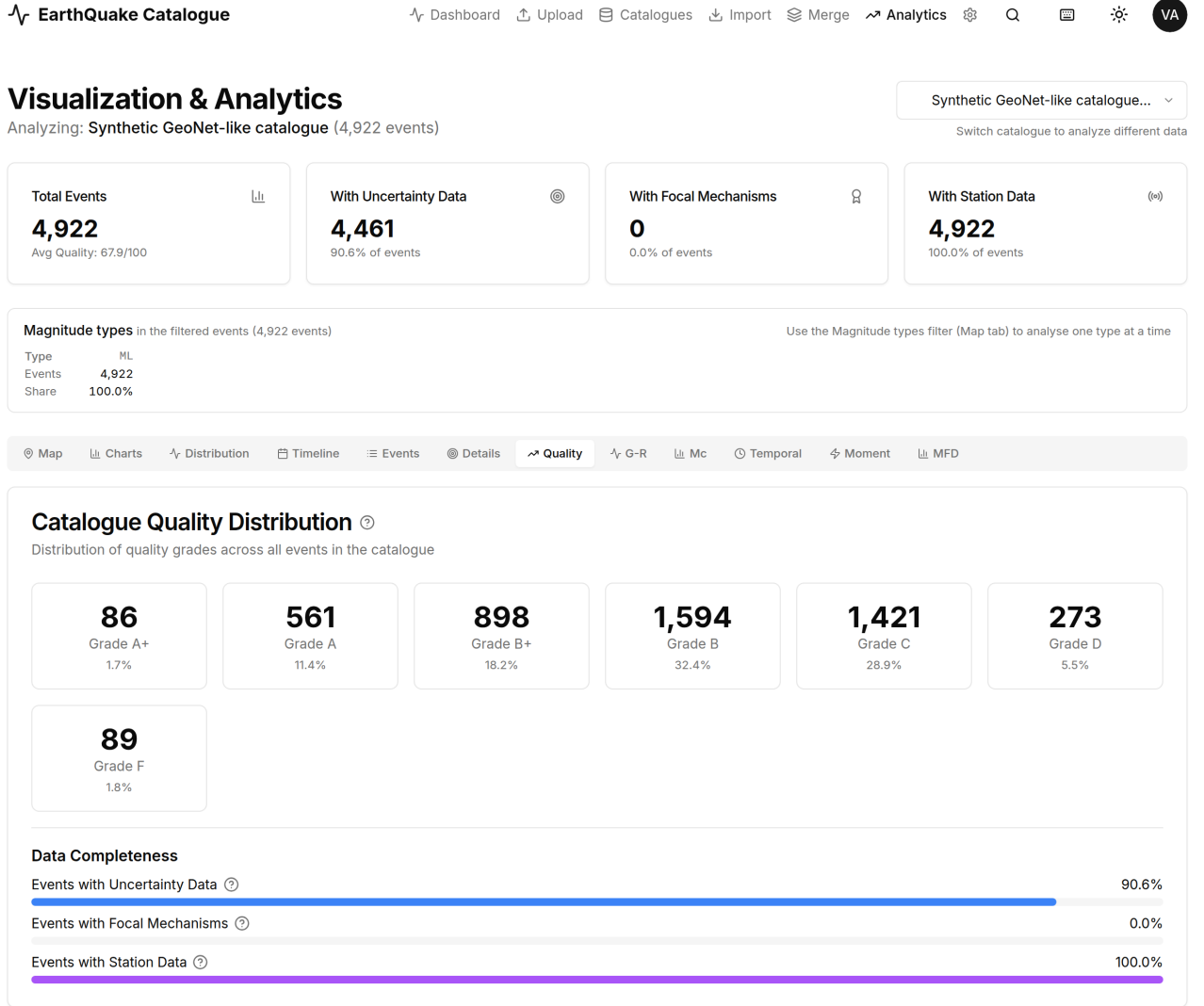


Figure S3: Catalogue-level quality summary for the same catalogue. The grade distribution and the mean score (67.9) are the five-dimension per-event index of equation 1 of the main paper, which scores location, network geometry, solution, magnitude, and evaluation status; every event is scored when it is stored and mapped to an A⁺–F grade. The synthetic reports carry the fields equation 1 reads (horizontal, depth and origin-time uncertainties, azimuthal gap, station and phase counts, RMS residual, magnitude uncertainty and station count, and an analyst-reviewed evaluation status), so the grades reflect the recording geometry: 86 events (1.7 %) reach A⁺, most events are graded B (32.4 %) or C (28.9 %), and the lowest grades go to offshore and small events recorded by few stations. The field-completeness bars are a separate catalogue-level metric: 90.6 % of the events have a location uncertainty in the sense of the “With Uncertainty Data” card (a latitude, longitude or depth uncertainty), the others having a fixed depth and only a horizontal uncertainty, and none has a focal mechanism, which equation 1 does not use.

References

- Båth, M. (1965). Lateral inhomogeneities of the upper mantle, *Tectonophysics* **2**, no. 6, 483–514.
- Omori, F. (1894). On the after-shocks of earthquakes, *Journal of the College of Science, Imperial University of Tokyo* **7**, 111–200.
- Shi, Y., and B. A. Bolt (1982). The standard error of the magnitude-frequency *b* value, *Bulletin of the*

- Seismological Society of America* **72**, no. 5, 1677–1687.
- Utsu, T. (1961). A statistical study on the occurrence of aftershocks, *Geophysical Magazine* **30**, 521–605.
- Utsu, T. (1970). Aftershocks and earthquake statistics (1): Some parameters which characterize an aftershock sequence and their interrelations, *Journal of the Faculty of Science, Hokkaido University, Series VII (Geophysics)* **3**, no. 3, 129–195.
- Wells, D. L., and K. J. Coppersmith (1994). New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement, *Bulletin of the Seismological Society of America* **84**, no. 4, 974–1002.
- Woessner, J., and S. Wiemer (2005). Assessing the quality of earthquake catalogues: Estimating the magnitude of completeness and its uncertainty, *Bulletin of the Seismological Society of America* **95**, no. 2, 684–698.